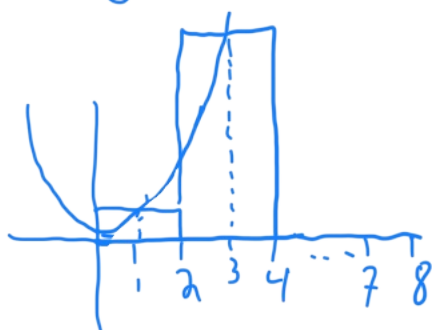


Name: solutions

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30 minutes. No aids (book, notes, calculator, internet, etc.) are permitted. Show all work and use proper notation for full credit. Put answers in reasonably-simplified form. 25 points possible.

1. [5 points] Estimate the value of $\int_0^8 x^2 dx$ using the midpoint rule with 4 sub-intervals.



$$\begin{aligned}
 \int_0^8 x^2 dx &\approx \frac{8}{4} \sum_{n=0}^3 (2n+1)^2 = \frac{1}{2} [1^2 + 3^2 + 5^2 + 7^2] \\
 &= \frac{1}{2} [10 + 25 + 49] \\
 &= \frac{1}{2} [84] \\
 &= \boxed{42}
 \end{aligned}$$

2. [12 points] Compute the following improper integrals, or show that they diverge. Use appropriate limit notation for improper integrals.

a. $\int_{-\infty}^0 x e^{3x} dx = \lim_{b \rightarrow -\infty} \int_b^0 x e^{3x} dx$

$u = x \quad v = \frac{1}{3} e^{3x}$
 $du = dx \quad dv = e^{3x} dx$

$$= \lim_{b \rightarrow -\infty} \left[\frac{x}{3} e^{3x} \right]_b^0 + \frac{1}{3} \int_b^0 e^{3x} dx$$

$$= \lim_{b \rightarrow -\infty} 0 - \frac{b}{3} e^{3b} + \frac{1}{9} [e^{3x}]_b^0$$

$$= 0 + \frac{1}{9} \lim_{b \rightarrow -\infty} (e^0 - e^{3b}) = \boxed{-\frac{1}{9}}$$

b. $\int_0^{\infty} \sin \theta d\theta = \lim_{b \rightarrow \infty} \int_0^b \sin \theta d\theta = \lim_{b \rightarrow \infty} [-\cos \theta]_0^b$

$$= \lim_{b \rightarrow \infty} -\cos b - (-\cos 0)$$

$$= 1 - \lim_{b \rightarrow \infty} \cos b$$

(DNE)

diverges

c. $\int_1^4 \frac{1}{\sqrt{4-x}} dx = \lim_{b \rightarrow 4^-} \int_1^b \frac{1}{\sqrt{4-x}} dx = \lim_{b \rightarrow 0^+} \int_3^b u^{-1/2} du$

$u = 4-x$
 $du = -dx$

$u(1) = 3$
 $u(4) = 0$

$$= \lim_{b \rightarrow 0^+} \int_b^3 u^{-1/2} du$$

$$= \lim_{b \rightarrow 0^+} [2\sqrt{u}]_b^3$$

$$= \lim_{b \rightarrow 0^+} 2\sqrt{3} - 2\sqrt{u}$$

$$= 2\sqrt{3} - 0$$

$$= \boxed{2\sqrt{3}}$$

3. [3 points] Find a formula for the n th term a_n of the sequence whose first several terms are

0, 1, 3, 7, 15, 31, 63, 127, 255, ...

$$\begin{aligned} 1+1 &= 2 = 2^1 \\ 3+1 &= 4 = 2^2 \\ 7+1 &= 8 = 2^3 \\ 15+1 &= 16 = 2^4 \\ &\vdots \end{aligned}$$

$$a_n + 1 = 2^n$$

$$a_n = 2^n - 1, \quad n = 0, 1, 2, \dots$$

$$2^0 - 1 = 1 - 1 = 0 \quad \checkmark$$

4. [6 points] For each sequence, find the limit if it converges, or show that the sequence diverges. Indicate any places where you use L'Hôpital's rule.

a. $\frac{\sqrt{n}}{\sqrt{n}+1}$

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n}+1} \stackrel{\text{L'H}}{=} \lim_{n \rightarrow \infty} \frac{\frac{1}{2\sqrt{n}}}{\frac{1}{2\sqrt{n}}} = \boxed{1}$$

b. $a_n = \frac{2^n + 3^n}{4^n}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2^n + 3^n}{4^n} &= \lim_{n \rightarrow \infty} \left(\left(\frac{2}{4} \right)^n + \left(\frac{3}{4} \right)^n \right) \\ &= \lim_{n \rightarrow \infty} \left(\left(\frac{1}{2} \right)^n + \left(\frac{3}{4} \right)^n \right) \\ &= 0 + 0 \\ &= \boxed{0} \end{aligned}$$

Extra Credit. [1 point] Suppose we use the n -subinterval midpoint and trapezoid rules on an integral $\int_a^b f(x) dx$. Assume that f is continuous, and that $f''(x) > 0$ (as in Problem 1). We get results M_n and T_n , and suppose we also have the exact value E . Which of the numbers M_n, T_n, E is largest? Which number is smallest?

$f''(x) > 0 \Rightarrow f$ is concave up on $[a, b]$



Midpoint underestimates



Trapezoid overestimates

$$M_n \leq E \leq T_n$$

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